

PRIYANSHU RANJAN

+91 8084344882

priyanshuranjan7856@gmail.com

LinkedIn

GitHub

LeetCode

CodeChef

Education

Birla Institute of Technology, Mesra

B.Tech in Computer Science & Engineering (AI & ML)

2023 – 2027

CGPA: 8.12/10

Surendranath Centenary School, Ranchi

Senior Secondary (CBSE) – 12th Grade

2023

Percentage: 89%

Surendranath Centenary School, Ranchi

Secondary (CBSE) – 10th Grade

2021

Percentage: 96%

Experience

Research Intern

May 2026 – present

BIT Mesra, Ranchi

- Developed a **CNN-Transformer-based** model for pre-apnea ECG classification on the **PhysioNet Apnea-ECG** dataset, achieving **87%+ accuracy** and **92%+ sensitivity** using 5-fold cross-validation.
- Designed a preprocessing pipeline with sliding-window segmentation, Z-score normalization, and class balancing to predict apnea events up to **45 seconds before onset** from single-lead ECG signals.
- Improved training efficiency by **~2×** using mixed-precision training and early stopping on GPU.

Projects

SyncSpace - Real-Time Collaborative Whiteboard:

[Live](#) [GitHub](#)

- Built a full-stack collaborative whiteboard with instant multi-user synchronization using **Socket.io**, **React**, and **Node.js**.
- Integrated **Groq AI (LLaMA 3.3 70B)** to generate architecture diagrams from natural language prompts, converting AI-generated JSON into **Rough.js** canvas elements in real time.
- Introduced **Redis** caching for canvas elements reducing repeated **MongoDB** reads by **99%**, serving returning users in under **5ms** vs **300ms**.
- Implemented **JWT authentication** and **socket-level** authorization, ensuring only authorized users can access or edit shared boards.
- Technologies: **React.js**, **Node.js**, **Express.js**, **Socket.io**, **MongoDB**, **Redis**, **Rough.js**, **Groq API**, **JWT**

CloudIT - Cloud-Based File Sharing :

[Github](#)

- Built a full-stack file-sharing platform supporting secure authentication, permission-based file access, and cloud-based file storage using **Cloudinary**.
- Integrated **JWT**-based authentication with access and refresh tokens, **Multer**-based file uploads, Cloudinary integration, and permission-based access file authorization to ensure secure file sharing.
- Added **Redis**-backend rate limiting to restrict login attempts and file uploads per user.
- Technologies: **React.js**, **Express.js**, **JavaScript**, **JWT**, **Bcrypt**, **Cloudinary**, **Multer**, **MongoDB**

Skills

- Languages:** C++, Python, JavaScript, HTML
- Frameworks & Libraries:** Node.js, Express.js, ReactJS, Tailwind CSS
- Tools/Platforms:** Docker, Postman, GitHub, MongoDB
- Relevant Coursework:** Data Structures, Computer Networks, Operating Systems, Object-Oriented Programming, Database Management Systems

Competitions & Achievements

- Solved **1000+ DSA problems** on **LeetCode**, **CodeChef** and **Codeforces**, achieving a LeetCode max rating of **1754**. [\[Link\]](#)
- Rated **2 Star** on CodeChef, achieving a global rank of **337** in **STARTERS 237**. [\[Link\]](#)
- Secured a **Top 5** position in **HackQuest'25**, the flagship hackathon of **Pantheon**, BIT Mesra's annual techfest.
- Secured a global rank of **3877** in a **LeetCode Weekly Contest** among **27,000+ participants**.

Honours and Awards

- Secured **State Rank 09** in the prestigious **National Talent Search Examination (NTSE)**.
- Achieved **All India Rank 03** ,in Abacus Prodigy Competition , declared **State Champion** and **City Topper** among **10K+ students** nationwide.
- Secured **All India Rank under 50** in a **National-level Painting Contest** organized by **HDFC**.

Position of Responsibility

- Executive Member-IEEE BIT Mesra:** Conducted a **DSA workshop for 100+ students**, introducing **competitive programming** and **data structures fundamentals**.